

Template Week 3 – Hardware

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Assignment 3.1: Examine your phone

What processor is in your phone?

Ik heb de iPhone 16, en die heeft een Apple A18 processor Apple Silicon

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

Deze processor gebruikt de ARM architectuur oftewel ARM64/AArch64

How much RAM is in it?

8 GB RAM

How much storage does your phone have?

256 GB

What operating system is running on your phone?

iOS 18

Approximately how many applications do you have installed?

Ik heb ongeveer 50/60

Which application do you use the most?

Snapchat

Can your phone be charged with what type of plug?

Ik gebruik de USB-C om op te laden maar het kan ook draadloos

Which I/O ports can you visually see on your phone?

USB-C poort, speakergrill, microfoon, SIM-slot, knoppen.

Assignment 3.2: Examine your laptop

What processor is in your laptop?

Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz (2.11 GHz)

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

64-bits besturingssysteem, x64-processor

How much RAM is in it?

8,00 GB

How much storage does your laptop have?

256 GB

Which operating system is running on your laptop?

Windows 11 Pro

Approximately how many applications do you have installed?

Ongeveer 40 gok ik

Which application do you use the most?

word

Can your laptop be charged with what type of plug?

Een ronde barrel-plug

Which I/O ports can you visually see on your laptop?

USB 3.1 Type-A poorten (2x)

USB 2.0 Type-A poort (1x)

HDMI-poort

RJ-45 Ethernet-poort

SD-kaartlezer

3.5mm audio jack

Oplaadpoort

Assignment 3.3: Power to the laptop

What is the input voltage?

100 – 249 V

What is the output voltage?

19.5 v dc

How many watts can your power adapter deliver?

65 watt (Ik gok even op mijn laptop dit alles maar er bestaan ook nog andere varianten met 45 watt)

Is the input voltage AC or DC?

Ac -= wisselspanning uit het stopcontact

Is the output voltage AC or DC?

Dc = gelijkspanning.

AC/DC what is that?

Ac:

Spanning wisselt heletijd van plus naar min

Komt uit stopcontact

Geschikt voor langere afstanden

Dc:

Spanning blijft heletijd positief

Word gebruikt door elektronica.

If you reverse the polarity of the output voltage, is that bad for your laptop?

Ja dat is heel slecht, waarom dat zo is omdat de laptop verwacht vanuit buitenaf dat de plug in negatief is en de binnenkant positief. Als je dat zou omdraaien loopt de stroom de verkeerde kant op. Dit kan daardoor onderdelen beschadigen.

You forgot your power adapter, your laptop normally needs 15 watts. You will be loaned a power adapter that can deliver 50 watts. Voltage, polarity, etc. are all the same compared to the original power adapter. You can connect the borrowed power adapter to your laptop. What will happen? Also explain why you think that.

Mijn laptop zou normaal werken dat komt omdat de laptop zelf bepaald hoeveel stroom hij nodig heeft. Hij pakt wat hij vraagt. Maar belangrijk is zolang spanning en polariteit het zelfde zijn, is het veilig.

Assignment 3.4: Build your dream PC

Screenshots PC configuration + motivation:

https://nl.pcpartpicker.com/list/Vi6dJ

Compatibility: See details below

Component	Selection	Base	Price	Shipping	Tax	Availability	Price	Where
CPU	Intel Core i9-12900K 3.2 GHz 16-Core Processor	€345.00	—	FREE	—	In stock	€345.00	AZERTY
CPU Cooler	ARCTIC Liquid Freezer III Pro 360 77 CFM Liquid CPU Cooler	€54.00	—	Prime	—	In stock	€54.99	amazon.nl
Motherboard	MSI PRO B660M-A WIFI DDR4 Micro ATX LGA1700 Motherboard	—	—	—	—	No Price Available	—	—
Memory	Corsair Vengeance LPX 32 GB (2 x 16 GB) DDR4-3200 CL16 Memory	€219.00	—	FREE	—	In stock	€219.90	amazon.nl
Storage	Samsung 990 EVO Plus 2 TB M.2-2280 PCIe 5.0 X2 NVMe Solid State Drive	€233.00	—	Prime	—	Available soon	€233.00	—
Video Card	Gigabyte WINDFORCE OC 0FF GeForce RTX 5070 12 GB Video Card	€519.00	—	FREE	—	In stock	€519.00	—
Case	Montech XR ATX Mid Tower Case	€59.00	—	FREE	—	In stock	€59.90	—
Power Supply	Corsair SF1000 (2024) 1000 W 80+ Platinum Certified Fully Modular SFX Power Supply	€224.00	—	Prime	—	In stock	€224.00	—
Operating System	Microsoft Windows 11 Home OEM - DVD 64-bit	€126.00	—	FREE	—	In stock	€126.00	—
Monitor	AOC Agon PRO AG276QZD2 26.7" 2560 x 1440 240 Hz Monitor	€480.00	—	FREE	—	In stock	€480.00	—

Expansion Cards / Networking: Sound Cards, Wi-Fi Network Adapters, Wireless Network Adapters

Peripherals: Headphones, Keyboards, Mice, Speakers, Webcams

Accessories / Other: Case Accessories, Case Fans, Fan Controllers, Thermal Compound, External Storage, Optical Drives, UPS Systems

Total: €2395.79

Buy from Amazon Netherlands

fio

Bestand Bewerken Weergeven

fiona lage Venterink

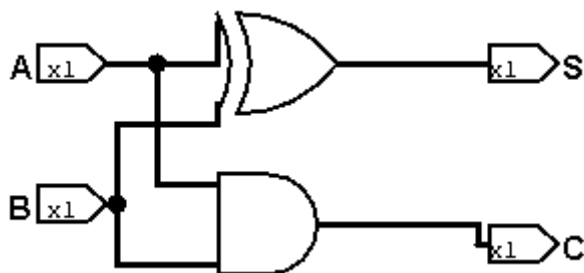
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Assignment 3.5: Adders

Complete the **half adder**, **full adder** and **4-bit adder** assignment as described in the PowerPoint slides of week 3 in Logisim. Save the chip design and also export three PNG pictures of the separate finished designs. See the PowerPoint slides of week 3.

Paste the three exported PNG pictures in here.



Ready? Save this file and export it as a pdf file with the name: **week3.pdf**